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Preconditioned GMRES methods for discretization equation of nonsymmetric and indefinite elliptic problem

Sheng Zhang

Institute of Computational Mathematics and Scientific/Engineering Computing, Academy of Mathematics and Systems Science, Chinese Academy of Science, P.O. Box 2719, Beijing 100080, People's Republic of China

Abstract

In this paper, a class of preconditioners for GMRES method are proposed for solving linear systems arising from discretization second-order nonsymmetric and indefinite elliptic problems by finite element method. The convergence of GMRES method is verified and the rate of convergence is shown. The generality of our result enables us to apply not only any known preconditioners designed for symmetric positive definite problems, but also many other preconditioners based on original operator to nonsymmetric and indefinite problems without losing optimality.
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1. Introduction

In this paper, we consider solving nonsymmetric and indefinite linear systems governed by a symmetric positive definite operator, which typically arise from discretization second-order nonsymmetric and indefinite elliptic problems by finite element method.

E-mail address: shzhang2000@263.sina.com (S. Zhang).

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In [11], Xu and Cai introduced a class of preconditioned GMRES methods for this kind of linear systems. The preconditioners they presented are constructed with the coarse-mesh solver plus any other preconditioners designed for the symmetric positive definite part of the original operator, or in other words, the corresponding symmetric positive definite problem. Also, they show that, following their approach of construction, optimal preconditioners for nonsymmetric and indefinite problem can be obtained from the coarse-mesh solver plus any optimal preconditioners for the corresponding symmetric positive definite problem. On the other hand, in [8], Kwak proposed GMRES method preconditioned with multigrid methods for the original problem for this kind of linear systems. The convergence of this type of preconditioned GMRES method is obtained and the corresponding rate of convergence is shown. It is notable that some preconditioners used in [8] are based on the original operator and are not included in the framework of Xu and Cai [11]. So, it is natural to consider the following question: is it possible to present a more general framework for constructing preconditioners, in which all the preconditioners proposed by Kwak [8] and Xu and Cai [11] and many other preconditioners are included, for GMRES method in order to solve this kind of linear systems? This is an interesting and important question both in theory and in practice. And this is exactly what we are going to talk about.

In the following sections, a framework of constructing preconditioners is proposed. The generality of our theory allows us to construct preconditioners not only from any known preconditioners which originally designed for the corresponding symmetric positive definite problem, but also from many other preconditioners based on original problem without losing optimality. All the preconditioners proposed by Kwak [8] and Xu and Cai [11] can be included in our framework and many other precondtioners can be obtained. In this sense, the class of preconditioners we construct here are more general.

The remainder of the paper is organized as follows: in Section 2, we introduce our model problem and some notations; in Section 3, preconditioners are constructed and analysis of convergence is carried out and in Section 4, some concrete examples are presented.

2. Model problem and notations

In this section, our model problem and some notations are introduced.

We adopt the notations for Sobolev space and their norms as presented in [4,5] and consider the following second-order elliptic boundary value problem:

$$\begin{cases} \nabla \cdot (-A(x) \cdot \nabla u + b(x)u) + q(x)u = f(x) & \text{in } \Omega, \\ u = 0 & \text{on } \partial\Omega, \end{cases} \tag{2.1}$$

where Ω is a polygonal domain (possibly nonconvex) in R^2, the matrix of coefficients $A := (a_{ij})_{i,j=1}^2$, $a_{ij} = a_{ji}$, $a_{ij} \in W^{1,\infty}(\Omega)$ for $i, j = 1, 2$, is symmetric and uniformly positive definite for all $x \in \Omega$. The vector of coefficients $b := (b_1, b_2)^{\mathrm{T}}$, b_i, $i = 1, 2$ is continuously differentiable on $\overline{\Omega}$, $q \in L^\infty(\Omega)$ is nonnegative and $f \in L^2(\Omega)$.

For $w, v \in H_0^1(\Omega)$, we define

$$a^{(2)}(w,v) = \int_\Omega (A\nabla w)\cdot\nabla v\,\mathrm{d}x, \quad a^{(1)}(w,v) = \int_\Omega (b\cdot\nabla v)w\,\mathrm{d}x,$$

$$a^{(0)}(w,v) = \int_\Omega qwv\,\mathrm{d}x, \quad a(w,v) = a^{(2)}(w,v) + a^{(1)}(w,v) + a^{(0)}(w,v),$$

$$\hat{a}(w,v) = a^{(2)}(w,v) + a^{(0)}(w,v), \quad (f,v) = \int_\Omega fv\,\mathrm{d}x.$$

Then the variational form for the model problem (2.1) is: find $u \in H_0^1(\Omega)$ such that

$$a(u,v) = (f,v) \quad \forall v \in H_0^1(\Omega).$$

From Green formula, it follows that the form $a^{(1)}(\cdot,\cdot)$ satisfies the following inequalities:

$$|a^{(1)}(w,v)| \leqslant c\|w\|_1\|v\|_0 \quad \text{and} \quad |a^{(1)}(w,v)| \leqslant c\|w\|_0\|v\|_1, \tag{2.2}$$

here and hereafter, c or C with or without subscript denotes a generic positive constant, which can take different values in different occurrences but always be independent of the meshsize and the number of levels.

By the assumptions on the coefficients appearing in the definition of $\hat{a}(\cdot,\cdot)$, it follows that for $v \in H_0^1(\Omega)$, there exist c and C such that

$$c\|v\|_1 \leqslant (\hat{a}(v,v))^{1/2} \leqslant C\|v\|_1.$$

The norm induced by the bilinear form $\hat{a}(\cdot,\cdot)$ is denoted as $\|v\|_{\hat{a}} = \hat{a}(v,v)^{1/2}$.

Next, two levels of triangulation are introduced. We first partition Ω into substructures $\{\Omega_i\}$, which provide a regular coarse mesh finite element triangulation $\mathscr{T}_H$ of Ω (cf. [4,5]). Define a mesh parameter $H = \max_i\{H_i\}$, where H_i is the diameter of Ω_i. Then, we divide each Ω_i into smaller elements, denoted by $\{\Delta_i^j, j = 1, 2, \ldots\}$, which form a regular fine mesh finite element triangulation $\mathscr{T}_h$ of Ω with the mesh parameter $h = \max_{i,j}\{h_i^j\}$, where h_i^j is the diameter of Δ_i^j. Let U^H and U^h be the continuous, piecewise linear finite element space with respect to the H-level triangulation $\mathscr{T}_H$ and h-level triangulation $\mathscr{T}_h$ respectively.

Then the variational problem associated with the finite element approximation on h-level is: find $u_h \in U^h$ such that

$$a(u_h,v) = (f,v) \quad \forall v \in U^h. \tag{2.3}$$

The corresponding error estimate is given by the following theorem (cf. [10]).

Theorem 1. *Let u and u_h be the solution of* (2.1) *and* (2.3), *respectively, then given* $\varepsilon > 0$, *there exists an* $H_0(\varepsilon) > 0$ *such that for all* $H \in (0, H_0]$

$$\|u - u_h\|_1 \leqslant \varepsilon \|f\|_0. \tag{2.4}$$

3. Preconditioners and analysis of convergence

In this section, preconditioners are constructed and the analysis of convergence for the corresponding preconditioned GMRES methods is carried out.

First, we define operators $Q_0 : U^h \mapsto U^H$, $P_0 : U^h \mapsto U^H$, $\hat{A}_h : U^h \mapsto U^h$, $A_h : U^h \mapsto U^h$ and $A_H : U^H \mapsto U^H$ as following:

$$(Q_0 w, v) = (w, v) \quad \forall w \in U^h, \ v \in U^H,$$

$$a(P_0 w, v) = a(w, v) \quad \forall w \in U^h, \ v \in U^H,$$

$$(\hat{A}_h w, v) = \hat{a}(w, v) \quad \forall w, v \in U^h,$$

$$(A_h w, v) = a(w, v) \quad \forall w, v \in U^h,$$

$$(A_H w, v) = a(w, v) \quad \forall w, v \in U^H.$$

It is well known that the operator P_0 has the following property (cf. [10]).

Lemma 2. *Given* $\varepsilon > 0$, *there exists an* $H_1(\varepsilon) > 0$ *such that if* $H \in (0, H_1]$, *then the following inequalities hold for all* $w \in U^h$

$$\|(I - P_0)w\|_0 \leqslant \varepsilon \|(I - P_0)w\|_1, \tag{3.1}$$

$$\|P_0 w\|_1 \leqslant C \|w\|_1. \tag{3.2}$$

With Lemma 2, we can obtain the following Lemma.

Lemma 3. *For all* $w \in U^h$, *the following inequality holds*:

$$\|P_0 w\|_{\hat{a}}^2 \leqslant \hat{a}(P_0 w, w) + C\varepsilon \|w\|_{\hat{a}}^2.$$

Proof. From the definition of P_0, (2.2), Lemma 2 and triangle inequality, it follows:

$$\begin{aligned}
\|P_0 w\|_{\hat{a}}^2 &= \hat{a}(P_0 w, P_0 w) = a(P_0 w, P_0 w) - a^{(1)}(P_0 w, P_0 w) \\
&= a(w, P_0 w) - a^{(1)}(P_0 w, P_0 w) = \hat{a}(w, P_0 w) + a^{(1)}((I - P_0)w, P_0 w) \\
&\leqslant \hat{a}(w, P_0 w) + c\|(I - P_0)w\|_0 \|P_0 w\|_{\hat{a}} \\
&\leqslant \hat{a}(w, P_0 w) + c\varepsilon\|(I - P_0)w\|_{\hat{a}} \|P_0 w\|_{\hat{a}} \\
&\leqslant \hat{a}(w, P_0 w) + c\varepsilon(\|w\|_{\hat{a}} + \|P_0 w\|_{\hat{a}})\|P_0 w\|_{\hat{a}} \\
&\leqslant \hat{a}(P_0 w, w) + C\varepsilon\|w\|_{\hat{a}}^2. \qquad \square
\end{aligned}$$

Next, we introduce our preconditioners. Let $\widetilde{B}$ be an operator which satisfies one of the following three conditions:

$$|\hat{a}((\widetilde{B} - \widehat{B})A_h w, v)| \leqslant C\|w\|_0 \|v\|_{\hat{a}} \quad \forall w, v \in U^h, \tag{3.3}$$

$$|\hat{a}((\widetilde{B} - \widehat{B})A_h w, v)| \leqslant C(\varepsilon + H)\|w\|_{\hat{a}} \|v\|_{\hat{a}} \quad \forall w, v \in U^h, \tag{3.4}$$

$$|\hat{a}((\widetilde{B} - \widehat{B})A_h w, v)| \leqslant C[\|w\|_0 + (\varepsilon + H)\|w\|_{\hat{a}}]\|v\|_{\hat{a}} \quad \forall w, v \in U^h, \tag{3.5}$$

where $\widehat{B}$ is a given symmetric positive defined preconditioner for $\hat{A}_h$. And define

$$B = (A_H)^{-1} Q_0 + \beta \widetilde{B}, \tag{3.6}$$

where β is a given positive constant depending on $\widetilde{B}$.

We use B as a preconditioner in cooperation with GMRES method (we refer to [6,9] for GMRES method in detail) to solve nonsymmetric and indefinite linear systems arising from discretization second-order nonsymmetric and indefinite elliptic problems by finite element method.

Suppose the linear system of equations to be solved is $Gx = g$, in which G is a linear operator defined on a finite-dimensional vector space $\widetilde{V}$ and g is a given vector in $\widetilde{V}$. Let $[\cdot, \cdot]$ be an inner product on $\widetilde{V}$ that induces the norm $|||\cdot|||$ and define

$$\alpha_0 = \inf_{w \neq 0} \frac{[w, Gw]}{[w, w]} \quad \text{and} \quad \alpha_1 = \sup_{w \neq 0} \frac{|||Gw|||}{|||w|||}.$$

Then the rate of the convergence of GMRES is given by the following theorem (cf. [6]).

Theorem 4. *If $\alpha_0 > 0$, then the GMRES method converges and after m steps, the norm of the residual is bounded by*

$$|||r_m||| \leqslant (1 - \alpha_0^2/\alpha_1^2)^{m/2} |||r_0|||,$$

where $r_m = g - Gx_m$.

In our case, $[\cdot, \cdot]$ is chosen to be $\hat{a}(\cdot, \cdot)$. And with this theorem, to verify the convergence of GMRES and present the corresponding rate of convergence,

we only needed to estimate the lower bound of α_0 and the upper bound of α_1 respectively. To this end, we present the following theorem.

Theorem 5. *There exist positive constants δ, γ, β and μ depending on λ_0 and λ_1 (the minimal and maximal eigenvalues of $\widehat{B}\hat{A}_h$, respectively), such that, for given $\varepsilon \in (0, \delta]$, there exists an constant $H_2 > 0$ such that, for all $H \in (0, H_2]$, the following inequalities hold for B given by* (3.6)

$$\hat{a}(BA_h w, w) \geqslant \gamma \|w\|_{\hat{a}}^2 \quad \forall w \in U^h, \tag{3.7}$$

$$\|BA_h w\|_{\hat{a}} \leqslant \mu \|w\|_{\hat{a}} \quad \forall w \in U^h. \tag{3.8}$$

Proof. From the definitions of P_0, A_H and A_h, it follows:

$$Q_0 A_h = A_H P_0$$

from which we obtain

$$BA_h = (A_H^{-1} Q_0 + \beta \widetilde{B}) A_h = P_0 + \beta \widetilde{B} A_h.$$

We assume that $\widetilde{B}$ satisfies condition (3.5) (the cases of $\widetilde{B}$ satisfies condition (3.3) or condition (3.4) can be treated similarly) and $H \in (0, H_1]$, where H_1 is given by Lemma 3. From (2.2) and the definitions of A_h and $\hat{A}_h$, it follows:

$$\begin{aligned} |\hat{a}(\widehat{B}(\hat{A}_h - A_h)w, v)| &= |\hat{a}(w, \widehat{B}\hat{A}_h v) - a(w, \widehat{B}\hat{A}_h v)| \\ &= |a^{(1)}(w, \widehat{B}\hat{A}_h v)| \leqslant C\|w\|_0 \|\widehat{B}\hat{A}_h v\|_{\hat{a}} \leqslant \lambda_1 \|w\|_0 \|v\|_{\hat{a}}. \end{aligned} \tag{3.9}$$

Then, by (3.5) and (3.9), we obtain

$$\begin{aligned} \|w\|_{\hat{a}}^2 &= \hat{a}(w, w) \leqslant \lambda_0^{-1} \hat{a}(\widehat{B}\hat{A}_h w, w) \\ &= \lambda_0^{-1}[\hat{a}(\widetilde{B}A_h w, w) + \hat{a}((\widehat{B} - \widetilde{B})A_h w, w) + \hat{a}(\widehat{B}(\hat{A}_h - A_h)w, w)] \\ &\leqslant \lambda_0^{-1}\{\hat{a}(\widetilde{B}A_h w, w) + C_1[\|w\|_0 + (\varepsilon + H)\|w\|_{\hat{a}}]\|w\|_{\hat{a}} + C_2\lambda_1\|w\|_0\|w\|_{\hat{a}}\} \\ &\leqslant \lambda_0^{-1}\hat{a}(\widetilde{B}A_h w, w) + C_3\lambda_0^{-1}(1 + \lambda_1)\|w\|_0\|w\|_{\hat{a}} + C_1(\varepsilon + H)\lambda_0^{-1}\|w\|_{\hat{a}}^2 \\ &\leqslant \lambda_0^{-1}\hat{a}(\widetilde{B}A_h w, w) + \frac{1}{2}C_3^2\lambda_0^{-2}(1 + \lambda_1)^2\|w\|_0^2 \\ &\quad + \frac{1}{2}\|w\|_{\hat{a}}^2 + C_1(\varepsilon + H)\lambda_0^{-1}\|w\|_{\hat{a}}^2, \end{aligned}$$

from which follows:

$$\|w\|_{\hat{a}}^2 \leqslant 2\lambda_0^{-1}\hat{a}(\widetilde{B}A_h w, w) + C_3^2\lambda_0^{-2}(1 + \lambda_1)^2\|w\|_0^2 + 2C_1(\varepsilon + H)\lambda_0^{-1}\|w\|_{\hat{a}}^2. \tag{3.10}$$

On the other hand, triangle inequality, Lemmas 2 and 3 yield

$$\begin{aligned}\|w\|_0^2 &\leqslant (\|P_0 w\|_0 + \|(I-P_0)w\|_0)^2\\ &\leqslant 2\|P_0 w\|_0^2 + 2\|(I-P_0)w\|_0^2\\ &\leqslant 2\|P_0 w\|_{\hat{a}}^2 + 2\varepsilon^2\|(I-P_0)w\|_{\hat{a}}^2\\ &\leqslant 2\|P_0 w\|_{\hat{a}}^2 + 2\varepsilon^2(\|w\|_{\hat{a}} + \|P_0 w\|_{\hat{a}})^2\\ &\leqslant 2\|P_0 w\|_{\hat{a}}^2 + C_4\varepsilon^2\|w\|_{\hat{a}}^2\\ &\leqslant 2\hat{a}(P_0 w, w) + C_5\varepsilon\|w\|_{\hat{a}}^2 + C_4\varepsilon^2\|w\|_{\hat{a}}^2\\ &= 2\hat{a}(P_0 w, w) + (C_5 + C_4\varepsilon)\varepsilon\|w\|_{\hat{a}}^2. \end{aligned} \tag{3.11}$$

So, from (3.10) and (3.11), it follows:

$$\begin{aligned}\|w\|_{\hat{a}}^2 \leqslant{}& 2\lambda_0^{-1}\hat{a}(\widetilde{B}A_h w, w) + 2C_3^2\lambda_0^{-2}(1+\lambda_1)^2\hat{a}(P_0 w, w)\\ &+ [C_3^2\lambda_0^{-2}(1+\lambda_1)^2(C_5 + C_4\varepsilon)\varepsilon + 2C_1\varepsilon\lambda_0^{-1}]\|w\|_{\hat{a}}^2 + 2C_1 H\lambda_0^{-1}\|w\|_{\hat{a}}^2. \end{aligned} \tag{3.12}$$

Let $\delta > 0$ be such that

$$C_3^2\lambda_0^{-2}(1+\lambda_1)^2(C_5 + C_4\delta)\delta + 2C_1\lambda_0^{-1}\delta = \frac{1}{2}.$$

Then, for $\varepsilon \in (0, \delta]$,

$$C_3^2\lambda_0^{-2}(1+\lambda_1)^2(C_5 + C_4\varepsilon)\varepsilon + 2C_1\lambda_0^{-1}\varepsilon \leqslant \frac{1}{2}$$

and it follows from (3.12) that

$$\begin{aligned}\|w\|_{\hat{a}}^2 \leqslant{}& 4\lambda_0^{-1}\hat{a}(\widetilde{B}A_h w, w) + 4C_3^2\lambda_0^{-2}(1+\lambda_1)^2\hat{a}(P_0 w, w)\\ &+ 4C_1 H\lambda_0^{-1}\|w\|_{\hat{a}}^2. \end{aligned} \tag{3.13}$$

Let $H_2 \leqslant H_1$ be a positive constant such that for all $H \in (0, H_2]$

$$4C_1 H\lambda_0^{-1} \leqslant \frac{1}{2}. \tag{3.14}$$

Then, for $H \in (0, H_2]$, (3.13) yields

$$\|w\|_{\hat{a}}^2 \leqslant 8\lambda_0^{-1}\hat{a}(\widetilde{B}A_h w, w) + 8C_3^2\lambda_0^{-2}(1+\lambda_1)^2\hat{a}(P_0 w, w)$$

from which follows (3.7) with

$$\gamma = \frac{\lambda_0^2}{8C_3^2(1+\lambda_1)^2} \quad \text{and} \quad \beta = \frac{\lambda_0}{C_3^2(1+\lambda_1)^2}.$$

On the other hand, taking $v = (\widetilde{B} - \widehat{B})A_h w$ and $v = \widehat{B}(\hat{A}_h - A_h)w$ in (3.5) and (3.9), respectively, yield

$$\|(\widetilde{B} - \widehat{B})A_h w\|_{\hat{a}} \leqslant c[\|w\|_0 + (\varepsilon + H)\|w\|_{\hat{a}}]$$

and

$$\|\widehat{B}(\hat{A}_h - A_h)w\|_{\hat{a}} \leqslant C\lambda_1 \|w\|_0.$$

With these inequalities, triangle inequality, Lemma 2 and (3.14), we obtain

$$\begin{aligned}
\|BA_h w\|_{\hat{a}} &\leqslant \|P_0 w\|_{\hat{a}} + \beta\|\widetilde{B}A_h w\|_{\hat{a}} \\
&\leqslant \|P_0 w\|_{\hat{a}} + \beta[\|(\widetilde{B} - \widehat{B})A_h w\|_{\hat{a}} + \|\widehat{B}(\hat{A}_h - A_h)w\|_{\hat{a}} + \|\widehat{B}\hat{A}_h w\|_{\hat{a}}] \\
&\leqslant \|P_0 w\|_{\hat{a}} + \beta\{c[\|w\|_0 + (\varepsilon + H)\|w\|_{\hat{a}}] + C\lambda_1\|w\|_0 + \lambda_1\|w\|_{\hat{a}}\} \\
&\leqslant \{C_6 + \beta[c(\varepsilon + H) + \lambda_1]\}\|w\|_{\hat{a}} + \beta(c + C\lambda_1)\|w\|_0 \\
&\leqslant \{C_6 + \beta[c(\varepsilon + H) + \lambda_1 + c + C\lambda_1]\}\|w\|_{\hat{a}} \\
&\leqslant \left\{C_6 + \beta\left[c\left(\varepsilon + \frac{\lambda_0}{8C_1}\right) + \lambda_1 + c + C\lambda_1\right]\right\}\|w\|_{\hat{a}} \\
&\leqslant [C_6 + C_7\beta(\varepsilon + \lambda_0 + \lambda_1 + 1)]\|w\|_{\hat{a}} \\
&\leqslant [C_6 + C_7\beta(\delta + 2\lambda_1 + 1)]\|w\|_{\hat{a}}
\end{aligned}$$

from which follows (3.8) with $\mu = C_6 + C_7\beta(\delta + 2\lambda_1 + 1)$. □

With Theorems 4 and 5, we arrive at the conclusion that for the preconditioners proposed by (3.6), the lower bound of α_0 and the upper bound of α_1 are given by

$$\alpha_0 > \gamma \quad \text{and} \quad \alpha_1 < \mu,$$

which guarantee the convergence of GMRES and present the corresponding rate of convergence.

Remark 6. Other Krylov-space methods like generalized conjugate residual algorithm introduce in [6] can also be used to solve the resulting discretization equation effectively instead of GMRES and these methods have the same rate of convergence as GMRES.

4. Some concrete examples

In this section, some concrete examples of this class of preconditioners are presented.

From the condition which $\widetilde{B}$ satisfies, it's obviously that any known preconditioners which originally designed for the corresponding symmetric posi-

tive definite problem can be used to construct preconditioners for nonsymmetric and indefinite problem without losing optimality, which means that all the preconditioners presented by Xu and Cai [11] are included in our framework. Moreover, in fact, the condition proposed for $\widetilde{B}$ is also satisfied by many preconditioners based on original problem. This implies that these preconditioners can also be used. In this sense, our framework is more general and enables us to obtain a much larger class of preconditioners for GMRES method in order to solve the linear systems arising from discretization second-order nonsymmetric and indefinite elliptic problems by finite element method. To illustrate this, we are going to show some concrete examples of preconditioners based on the original operator. In [8], Kwak presented GMRES method preconditioned with multigrid methods based on the original problem. Next, we show that this kind of preconditioned GMRES method can also be included in our framework.

First, we introduce some notations and theorems for multigrid method as preparation (we refer to [1,7] and the cited papers therein for multigrid method in detail).

To present multigrid method, some multilevel notations defined below are needed. For a given triangulation $\mathscr{T}_k$, $k = 0, 1, \ldots, J-1$, $\mathscr{T}_{k+1}$ is obtained by connecting the midpoints of triangles in $\mathscr{T}_k$, where $\mathscr{T}_0 = \mathscr{T}_H$ and $\mathscr{T}_J = \mathscr{T}_h$. The mesh parameters of $\mathscr{T}_k$, $k = 0, 1, \ldots, J$, are denoted as h_k, $k = 0, 1, \ldots, J$, respectively, with $h_0 = H$ and $h_J = h$. Let U_k, $k = 1, \ldots, J-1$, be the continuous, piecewise linear finite element spaces with respect to the triangulations $\mathscr{T}_k$ and $U_0 = U^H$, $U_J = U^h$. Define operators $A_k : U_k \mapsto U_k$, $\hat{A}_k : U_k \mapsto U_k$, $P_k : H_0^1(\Omega) \mapsto U_k$, $\hat{P}_k : H_0^1(\Omega) \mapsto U_k$ and $P_{k-1}^0 : U_k \mapsto U_{k-1}$ as following:

$$(A_k w, v) = a(w, v), \quad (\hat{A}_k w, v) = \hat{a}(w, v) \ \ \forall w, v \in U_k,$$

$$a(P_k w, v) = a(w, v), \quad \hat{a}(\hat{P}_k w, v) = \hat{a}(w, v) \ \ \forall v \in U_k,$$

$$(P_{k-1}^0 w, v) = (w, v) \quad \forall w \in U_k, \ \ v \in U_{k-1},$$

where $A_J = A_h$, $\hat{A}_J = \hat{A}_h$ and $A_0 = A_H$.

Let R_k be a certain smoother, then multigrid algorithm for nonsymmetric and indefinite problem can be described as follows:

Multigrid algorithm. Set $B_0(v, w) = A_0^{-1} w$. For $k > 0$ define B_k in terms of B_{k-1} as follows:

1. Define $x_k = v + R_k(w - A_k v)$.
2. Set $B_k(v, w) = x_k + q$, where q is defined by

$$q = B_{k-1}(0, P_{k-1}^0(w - A_k x_k)).$$

Define $K_k = I - R_k A_k$, $k = 0, 1, \ldots, J$, $T_k = R_k A_k P_k$ for $k > 0$ and $T_0 = P_0$. And let $E_J = I - B_J A_h$. Also, with $\hat{A}_k$, $\hat{P}_k$ and certain smoother $\widehat{R}_k$, the corresponding operators $\widehat{B}_J$, $\widehat{E}_J$, $\widehat{K}_k$ and $\widehat{T}_k$ for the symmetric positive definite problem can be defined similarly. Following [3], if the smoother $\widehat{R}_k$ satisfy conditions (C1) and (C2) given below.

(C1) There exists a constant C_R such that

$$\frac{(w,w)_k}{\lambda_k} \leqslant C_R(\overline{R}_k w, w)_k \quad \forall w \in U_k,$$

where $(w,v)_k = h_k^2 \sum_i w(x_i)v(x_i)$, the sum is taken over all nodes x_i of U_k, λ_k is the largest eigenvalue of $\hat{A}_k$, $\overline{R}_k = (I - \widehat{K}_k^* \widehat{K}_k)\hat{A}_k^{-1}$ and the superscript * denotes the adjoint with respect to the inner product $\hat{a}(\cdot,\cdot)$.

(C2) There exists a constant $\theta < 2$ independent of k satisfying

$$\hat{a}(\widehat{T}_k v, \widehat{T}_k v) \leqslant \theta \hat{a}(\widehat{T}_k v, v) \quad \forall v \in U_k.$$

Then, the following theorem holds (cf. [3]).

Theorem 7. *There exists a positive constant $\hat{\delta} < 1$ independent of J such that*

$$\hat{a}(\widehat{E}_J u, \widehat{E}_J u) \leqslant \hat{\delta}^2 \hat{a}(u,u) \quad \forall u \in U_J. \tag{4.1}$$

Also, for nonsymmetric and indefinite problem, it is well known that, if R_k satisfy certain condition, then the following theorem holds (cf. [2] for detail).

Theorem 8. *Given $\varepsilon > 0$, there exists a constant $H_3 > 0$ such that for $H \leqslant H_3$, the following inequalities hold*

$$\|E_J - \widehat{E}_J\|_{\hat{a}} \leqslant C(H + \varepsilon), \tag{4.2}$$

$$\hat{a}(E_J u, E_J u) \leqslant \tilde{\delta}^2 \hat{a}(u,u) \quad \forall u \in U_J. \tag{4.3}$$

where $\tilde{\delta} = \hat{\delta} + C(H + \varepsilon)$ and $\hat{\delta}$ is given by Theorem 7.

And, in [2], some R_k of this kind such as point, block Jacobi, Gauss–Seidel and so on are presented.

With these preparations, we can show the convergence of GMRES method preconditioned with multigrid method based on the original operator for nonsymmetric and indefinite problem in our framework now. To do this, we only need to verify that B_J satisfies one of (3.3), (3.4) and (3.5). This is done by the following theorem.

Theorem 9. *The following inequality holds for all* $w, v \in U^h$,

$$|\hat{a}((B_J - \widehat{B}_J)A_h w, v)| \leqslant C[\|w\|_0 + (\varepsilon + H)\|w\|_{\hat{a}}]\|v\|_{\hat{a}}. \tag{4.4}$$

Proof. From (2.2), triangle inequality, Theorems 7 and 8, it follows:

$$\begin{aligned}
|\hat{a}((B_J - \widehat{B}_J)A_h w, v)| &= |\hat{a}(B_J A_h w, v) - \hat{a}(\widehat{B}_J \hat{A}_h w, v) \\
&\quad + \hat{a}(\widehat{B}_J \hat{A}_h w, v) - \hat{a}(\widehat{B}_J A_h w, v)| \\
&= |\hat{a}((\widehat{E}_J - E_J)w, v) + \hat{a}(w, \widehat{B}_J \hat{A}_h v) - a(w, \widehat{B}_J \hat{A}_h v)| \\
&= |\hat{a}((\widehat{E}_J - E_J)w, v) - a^{(1)}(w, \widehat{B}_J \hat{A}_h v)| \\
&\leqslant \|(E_J - \widehat{E}_J)w\|_{\hat{a}}\|v\|_{\hat{a}} + C\|w\|_0\|\widehat{B}_J \hat{A}_h v\|_{\hat{a}} \\
&\leqslant \|(E_J - \widehat{E}_J)w\|_{\hat{a}}\|v\|_{\hat{a}} + C\|w\|_0(\|v\|_{\hat{a}} + \|\widehat{E}_J v\|_{\hat{a}}) \\
&\leqslant [\|(E_J - \widehat{E}_J)\|_{\hat{a}}\|w\|_{\hat{a}} + C\|w\|_0(1 + \hat{\delta})]\|v\|_{\hat{a}} \\
&\leqslant C[\|w\|_0 + (\varepsilon + H)\|w\|_{\hat{a}}]\|v\|_{\hat{a}}. \qquad \square
\end{aligned}$$

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